

EETS 7302 TCP/IP Network Administration

Session 10

Linux 2: Admin

Identifying Commands

- It is often useful to know exactly which of the four kinds of commands is being used and Linux provides a couple of ways to find out.
- **type**
 - The type command is a shell builtin that displays the kind of command the shell will execute, given a particular command name. It works like this:
 - `type command` where “command” is the name of the command you want to examine.

which

- **which**

- Sometimes there is more than one version of an executable program installed on a system. While this is not very common on desktop systems, it's not unusual on large servers. To determine the exact location of a given executable
- which only works for executable programs, not builtins nor aliases that are substitutes for actual executable programs.

help

- Getting Command Documentation:
 - With this knowledge of what a command is, we can now search for the documentation available for each kind of command.
- **help**
 - bash has a built-in help facility available for each of the shell builtins. To use it, type “help” followed by the name of the shell builtin. Optionally, you may add the -m option to change the format of the output.
 - **A note on notation:** When square brackets appear in the description of a command's syntax, they indicate optional items. A vertical bar character indicates mutually exclusive items. In the case of the cd command above:

man

- **man**

- Most executable programs intended for command line use provide a formal piece of documentation called a *manual* or *man page*. A special paging program called man is used to view them. It is used like this:
 - `man program`

Top command:

- one of the most frequently used commands in administrative jobs.
- top command displays processor activity of your Linux box and also displays tasks managed by kernel in real-time. It'll show processor and memory are being used and other information like running processes. This may help you to take correct action. top command found in UNIX-like operating systems.

command	description
Space	Immediately refresh the display
h	Display a help screen
k	Kill a process. You are prompted for the process ID and the signal to send to it.
n	Change the number of processes displayed. You are prompted to enter the number.
u	Sort by user.
M	Sort by memory usage.
P	Sort by CPU usage.

yum utility

- To manage and let's say if we want to keep the software updated we can use yum utility to modify our system.
- There are basically four ways to use yum utility
 - To install new software from package repositories
 - To install new software from an individual package file
 - To update existing software on your system
 - To remove unwanted software from your system

yum utility

- Setting UP a yum repository:
 - To set up a repository, follow these steps:
 - Install the createrepo package.
 - Copy all the packages into one directory (for example, /mnt/local_repo).
 - Run createrepo on that directory (for example, createrepo /mnt/local_repo). This will create the necessary metadata for your yum repository.

yum utility

- Yum commands:
 - yum commands are typically run as yum **<command>** **<package name/s>**.
 - By default, yum will automatically attempt to check all configured repositories to resolve all package dependencies during an installation/upgrade.
 - The following is a list of the most commonly-used yum commands. For a complete list of available yumcommands, refer to man yum.

yum utility

- yum install **<package name/s>**
 - Used to install the latest version of a package or group of packages. If no package matches the specified package name(s), they are assumed to be a shell glob, and any matches are then installed.
- yum update **<package name/s>**
 - Used to update the specified packages to the latest available version. If no package name/s are specified, then yum will attempt to update all installed packages.

yum utility

- If the --obsoletes option is used (i.e. yum --obsoletes **<package name/s>**, yum will process obsolete packages. As such, packages that are obsoleted accross updates will be removed and replaced accordingly.
- yum check-update
 - This command allows you to determine whether any updates are available for your installed packages. Yum returns a list of all package updates from all repositories if any are available.

yum utility

- yum remove **<package name/s>**
 - Used to remove specified packages, along with any other packages dependent on the packages being removed.
- yum provides **<file name>**
 - Used to determine which packages provide a specific file or feature.
- yum search **<keyword>**
 - This command is used to find any packages containing the specified keyword in the description, summary, packager and package name fields of RPMs in all repositories.

yum utility

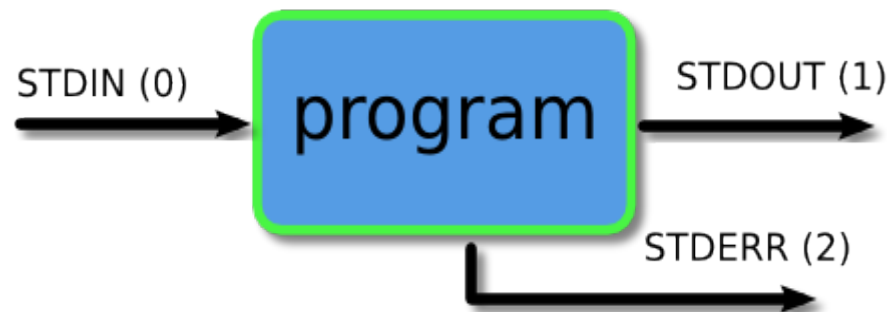
- yum localinstall <***absolute path to package name/s***>
 - Used when using yum to install a package located locally in the machine.

Redirection & piping

- Anything you type on your computer, the results are flashed on your screen.
- The screen is the default output for your computer.
- Similarly on linux, when you are working on a command line, there are some destinations decided by the interpreter or the shell from where it accepts the inputs, it sends the error messages and flashes the results on the screen.
- There are 3 types of input and output techniques used in linux.
 - The input to the shell is the standard input abbreviated as STDIN.
 - Output of the results produced by the shell are called STDOUT
 - The error messages produced by the shell is called STDERR.

Redirection & piping

- Any process that is running has 3 files namely file 0, file 1 and file 2.
- The standard input has the file descriptor as 0, standard output has the file descriptor as 1 and the standard error has the file descriptor as 2.
- Therefore the output can either be the correct output or can be an error message which will be returned to the screen.
- With the help of pipes and redirections we can combine or append outputs of many operations and perform the tasks in simple ways.



Redirections

- Redirections are described in arithmetic operational symbols like the greater than or less than signs. Refer the table below for more information.

Names	Abbreviation	File Description	Redirection symbols used
Standard input	STDIN	0	< or 0<
Standard output	STDOUT	1	> Or 1>
Standard error	STDERR	2	2>

Redirections

- Redirections are used for redirecting outputs
- The input to the shell is from the keyboard, the output and error messages are displayed on the screen.
- But there are many applications or processes that run in the background and not foreground of your screen. Their outputs are not flashed on the screen.

Redirections - example

```
[adminuser@localhost ~]$ ls
barry.txt  Documents  firstfile  not-here  result
blah.foo  Downloads  fool       ok.txt    Templates
bob        example.png Music      Pictures  video.mpg
Desktop    file1      myoutput   Public    Videos
[adminuser@localhost ~]$ ls > result
[adminuser@localhost ~]$ cat result
barry.txt
blah.foo
bob
Desktop
Documents
Downloads
example.png
file1
firstfile
fool
Music
myoutput
not-here
ok.txt
Pictures
Public
result
Templates
video.mpg
Videos
[adminuser@localhost ~]$
```

Piping: |

- Pipes are used to channel the output of 1 command as an input to the other command.
- As an admin, one will be using piping on a large scale. Using piping making navigation around linux quite easy and flexible.
- | is the symbol used for piping.
- Example:
 - `$ cat file1 | less`
 - cat command will display the contents of the file file1, the output which will be forwarded as input to the less command.
 - Less command will page the output and the user the user will be easily be able to navigate up and down through the page
-

System admin

- There are 2 types of users – Privileged and unprivileged.
- Privileged users are the ones who have default access to literally everything on the linux system. They also have access to the system hardware.
- Unprivileged users are the ones who do not have access to the system hardware.
- Privileged users are called the root users or the super users.
- Super users can perform all administration tasks. They have access to the shell configuration files, they can install software on the system, manage all the users on the linux systems, manage the disk space or partitioning of the disk devices.

System admin

- Any user account information is stored in the `/etc/passwd` file.
- This file is globally readable.
- The user account have the following categories-
- Username – There can be no spaces when you create a username.
- Password – stored in `/etc/shadow` file.

SU

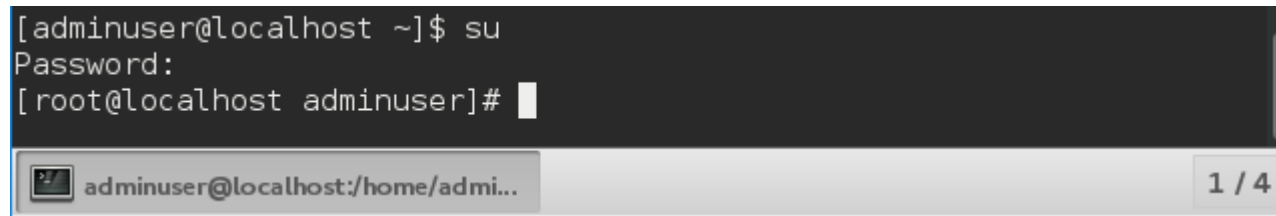
- The su command enables clients to open a terminal window, and from that terminal begin a sub shell in which the client has super user (root) role.
- To perform managerial undertakings, for example, you can sign in with an ordinary client record and sort *su* to open a root shell.



```
[adminuser@localhost ~]$ su
Password:
[root@localhost adminuser]#
```

The image shows a terminal window with a dark background. The first line shows the prompt [adminuser@localhost ~]\$ followed by the command su. The second line shows the prompt Password: with no visible input. The third line shows the prompt [root@localhost adminuser]# with a cursor. Below the terminal window is a light gray bar containing a small icon of a terminal window, the text adminuser@localhost:/home/admi..., and a page indicator 1 / 4.

- In the event that simply the command `su` is written, the username `root` is inferred. Be that as it may, `su` can be utilized to run undertakings as another client too.
- Sort `su bob` to open a subshell as the client `bob`, for instance. When utilizing `su` as a customary client, you are prompted for a password and in the wake of entering that you have procured the authority of that client.

A terminal window screenshot showing the execution of the 'su' command. The prompt is '[adminuser@localhost ~]\$' and the command entered is 'su'. The next line shows 'Password:' followed by a hidden password (represented by dots). The final line shows the prompt has changed to '[root@localhost adminuser]#', indicating a successful switch to the root user. The terminal window has a dark background and a light-colored title bar at the bottom showing the current directory and a page indicator '1 / 4'.

```
[adminuser@localhost ~]$ su
Password:
[root@localhost adminuser]#
```

sudo

- Rather than utilizing the root account, unprivileged clients/users can be designed for utilizing consents on particular tasks by utilizing *sudo* command.
- Ordinary clients or users have sudo benefits and to utilize these benefits, they will begin utilizing sudo. In this way, rather than utilizing commands like *useradd* as the root client, you utilize a privileged client account and use *sudo useradd* .

sudo	su
Need to enter user's password.	Need to enter the root user's password.
Lets you the access or permission for just one command.	Gives you privileges for the entire code.
Only allows certain people with certain authority privileges and not all.	Anyone who knows the root password has the privilege.

Admin: create user

- Username criteria-
 - Linux supports username upto 32 characters but its best advised to have 8.
 - Usernames are basically case sensitive.
 - Usernames do not contain spaces or special characters.
- Useradd command
- This command is used to add user to the system.

Format for the command is

```
$ useradd [options] username
```

Use man command to utilize options

Admin: create user

- Examples-

- 1. `useradd -c "bob marley" -m -s /bin/bash bob`

In the above example, a username= bob, shell= bash, and account is created for Bob Marley.

- 2. `useradd -m -u 1024 -G dancers sam`

In the above example, a user Sam is created who is member of the group dancers with a UID 1024 and a home directory is also added.

Note- Whenever a new user is created, you are always prompted to set a new password for that particular account. Type the password twice to confirm it.

Admin: create account

- For system account,
- Not all users on the system are humans and some accounts are run by the system itself.
- Let us create an account for a web server.
- `useradd -c "web server" -d /opt/server -r -s /sbin/nologin web`
- Note- In the above example, -d denotes the home directory for a system and -r creates a system account.

Admin: Delete user

- userdel command
- This command is used to delete a user from the system.
- Format –
- userdel [options] username

Admin: modify user accounts

- “usermod” command lets you modify the existing accounts of the users.
- `usermod [options] username`

Options	Function
-a	Is used with -G, adds users to supplemental groups.
-l (login name)	Changes the username
-L	Locks the account
-U	Unlocks the account
-m	Moves the directory
-s	Specifies the shell
-u	Modifies the UID

Admin: group info file

- /etc/group – This file contains all the group information.
- Contains 4 fields as follows-
- Groupname
- Password
- Group ID
- Members- Users who are members of that particular group as a secondary group.

```
[root@localhost adminuser]# cat /etc/group
root:x:0:
bin:x:1:
daemon:x:2:
sys:x:3:
adm:x:4:
tty:x:5:
disk:x:6:
lp:x:7:
mem:x:8:
kmem:x:9:
wheel:x:10:sneha,lisa
```

Admin: Create groups

- groupadd command
- Format-
- `groupadd [options] groupname`
- Lets take an example-

```
[root@localhost adminuser]# groupadd club
[root@localhost adminuser]# tail -1 /etc/group
club:x:1009:
[root@localhost adminuser]# groupadd -g 1234 dancers
[root@localhost adminuser]# tail -1 /etc/group
dancers:x:1234:
[root@localhost adminuser]# █
```

Lets break this down-

Line 1 – group club is created

Line 2- we confirm that by checking in the/etc/group file

Line 4 – group dancers is created with a GID 1024

Line 5- we confirm that by checking in /etc/group file.

Admin: delete groups

- groupdel command
- This command is used to delete a group.
- Format –
 - `groupdel [options] groupname`
- groupmod command
- This modifies the properties of all groups. It will allow the user to make modifications with the name or ID of the user.

Package Management

- What is a package?
- Package is nothing but a basic collection of files.
- Contains Data files and binary programs.
- Contains metadata files-

It consists of the following-

Package name and description

Version number

CPU architecture

Releases

Dependencies and many more.

Package Management

- Package managers- They perform the following functions-
Update, upgrade, delete, install the packages.
- They take charge of all the dependencies and resolve them.
- Maintain a record of what is installed, where it is installed, version number etc in the database.
- 2 most commonly used package manager formats are
 1. RPM
 2. DEB

Package Management

- Linux supports 3 methods of package installation on its system. They are as follows-
 - 1. Downloading the package directly from the internet or the web.
 - 2. Installing the packages from the software repository.
 - 3. Installing from the source codes. Here the codes are compiled to binary codes and then installed on the user's host.

Package Management

- Note- The best and the standard way of installing packages is by installing from the repository. It is the easiest method and also resolves dependencies.
- If there is no other way, only then go for other two methods.

Package Management

- RPM supported distributions-
- 1. Red Hat
- 2. CentOS
- 3. Fedora
- 4. Open SUSE
- 5. Scientific Linux

Package Management

- Debian supported distributions-
- 1. Ubuntu
- 2. Debian
- 3. Linux Mint
- 4. Kali

Package Management

RPM	DEBIAN
RPM does not auto configure or autostart services on the system.	Debian auto configures or autostarts services on the system.
File extension is .rpm	File extension is .deb
Creating repositories is easy task.	Creating repositories is difficult.
Creating packages is difficult.	Creating packages is easy.
Used on linux server installations.	Used on linux desktop installations.

yum

- yum command-
- If you want to make use of repositories, you need the yum command.
- need to know the correct name of the package.
- The yum search option helps a lot. it initially connects with the online repositories (which may pause for a moment), after which it downloads the latest archive metadata on the system machine.

Yum-

```
[root@localhost adminuser]# yum search bash
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
* base: centos.mirror.lstn.net
* extras: mirror.hmc.edu
* updates: mirror.clarkson.edu
===== N/S matched: bash =====
bash-completion.noarch : Programmable completion for Bash
bash-doc.x86_64 : Documentation files for bash
libguestfs-bash-completion.noarch : Bash tab-completion scripts for
                                         : libguestfs tools
pcp-pmda-bash.x86_64 : Performance Co-Pilot (PCP) metrics for the
                                         : Bash shell
bash.x86_64 : The GNU Bourne Again shell

Name and summary matches only, use "search all" for everything.
[root@localhost adminuser]#
```

- Note- In the example, the string bash is searched for. All the searches will be highlighted in bold.
- To access yum, one has to have root privileges.

Yum-

- But the search command is sometimes not sufficient. If you want to search for a particular file, give the command, *yum provide */filename* to see what package does that file belong to.

```
[root@localhost adminuser]# yum provides nmcli
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
 * base: centos.mirror.lstn.net
 * extras: centos.den.host-engine.com
 * updates: mirror.clarkson.edu
1:NetworkManager-1.8.0-9.el7.x86_64 : Network connection manager and user applications
Repo      : base
Matched from:
Filename  : /usr/bin/nmcli

1:NetworkManager-1.8.0-11.el7_4.x86_64 : Network connection manager and user applications
Repo      : updates
Matched from:
Filename  : /usr/bin/nmcli

1:NetworkManager-1.0.6-30.el7_2.x86_64 : Network connection manager and user applications
Repo      : @updates
Matched from:
Filename  : /bin/nmcli

1:NetworkManager-1.0.6-30.el7_2.x86_64 : Network connection manager and user applications
Repo      : @updates
Matched from:
Filename  : /usr/bin/nmcli
```

yum options-

options	Descriptions
install / localinstall	To install a package
remove	To remove a package
info	To give description of the package
search	Search for the package
provides	Provides the package the file belongs to
update	To update all packages
upgrade	To upgrade all packages
clean all	To remove the metadata
list all	Lists all installed packages

yum info –

```
[root@localhost adminuser]# yum info inkscape
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
* base: centos.mirror.lstn.net
* extras: mirror.atlantic.net
* updates: centos.unixheads.org
Available Packages
Name           : inkscape
Arch           : x86_64
Version        : 0.91
Release        : 2.el7
Size           : 10 M
Repo           : base/7/x86_64
Summary        : Vector-based drawing program using SVG
URL            : http://inkscape.sourceforge.net/
License        : GPLv2+
Description    : Inkscape is a vector graphics editor, with capabilities similar to
                : Illustrator, CorelDraw, or Xara X, using the W3C standard Scalable Vector
                : Graphics (SVG) file format. It is therefore a very useful tool for web
                : designers and as an interchange format for desktop publishing.
                :
                : Inkscape supports many advanced SVG features (markers, clones, alpha
                : blending, etc.) and great care is taken in designing a streamlined
                : interface. It is very easy to edit nodes, perform complex path operations,
                : trace bitmaps and much more.
```

yum install-

```
Verifying : inkscape-0.91-2.el7.x86_64 5/12
Verifying : 1:numpy-1.7.1-11.el7.x86_64 6/12
Verifying : lapack-3.4.2-8.el7.x86_64 7/12
Verifying : libquadmath-4.8.5-16.el7.x86_64 8/12
Verifying : gc-7.2d-7.el7.x86_64 9/12
Verifying : atlas-3.10.1-12.el7.x86_64 10/12
Verifying : python-nose-1.3.7-1.el7.noarch 11/12
Verifying : ImageMagick-c++-6.7.8.9-15.el7_2.x86_64 12/12

Installed:
  inkscape.x86_64 0:0.91-2.el7

Dependency Installed:
  ImageMagick-c++.x86_64 0:6.7.8.9-15.el7_2  atlas.x86_64 0:3.10.1-12.el7  blas.x86_64 0:3.4.2-8.el7  gc.x86_64 0:7.2d-7.el7  gsl.x86_64 0:1.15-13.el7  gtkspell.x86_64 0:2.0.16-8.el7
  lapack.x86_64 0:3.4.2-8.el7  libgfortran.x86_64 0:4.8.5-16.el7  libquadmath.x86_64 0:4.8.5-16.el7  numpy.x86_64 1:1.7.1-11.el7  python-nose.noarch 0:1.3.7-1.el7

Complete!
[root@localhost adminuser]#
```

- Whenever you install a package a prompt appears asking whether or not to install it. Type y for yes and n for no.

yum remove –

```
Removed:
  inkscape.x86_64 0:0.91-2.el7

Complete!
[root@localhost adminuser]#
```

- yum list installed-

```
nautilus-extensions.x86_64
nautilus-open-terminal.x86_64
nautilus-sendto.x86_64
ncompress.x86_64
ncurses.x86_64
ncurses-base.noarch
ncurses-libs.x86_64
neon.x86_64
net-snmp-libs.x86_64
net-tools.x86_64
netcf-libs.x86_64
nettle.x86_64
newt.x86_64
newt-python.x86_64
nfs-utils.x86_64
nfs4-acl-tools.x86_64
nhn-nanum-fonts-common.noarch
nhn-nanum-gothic-fonts.noarch
nm-connection-editor.x86_64
nmap-ncat.x86_64
nspr.x86_64
nss.x86_64
nss-softokn.x86_64
nss-softokn-freebl.x86_64
nss-sysinit.x86_64
nss-tools.x86_64
nss-util.x86_64
ntp.x86_64
```

yum group command-

- Package groups are efficient to use instead of specific individual package.
- A package group is defined in the repository, and yum offers certain group commands to work smoothly with these groups.
- For an overview or information of all current groups on the system,
- use `yum groups list` .

```
[root@localhost adminuser]# yum groups list
Loaded plugins: fastestmirror, langpacks
There is no installed groups file.
Maybe run: yum groups mark convert (see man yum)
Loading mirror speeds from cached hostfile
* base: centos.mirror.lstn.net
* extras: centos.den.host-engine.com
* updates: centos.den.host-engine.com
Available Environment Groups:
  Minimal Install
  Compute Node
  Infrastructure Server
  File and Print Server
  Basic Web Server
  Virtualization Host
  Server with GUI
  GNOME Desktop
  KDE Plasma Workspaces
  Development and Creative Workstation
Available Groups:
  Compatibility Libraries
  Console Internet Tools
  Development Tools
  Graphical Administration Tools
  Legacy UNIX Compatibility
  Scientific Support
  Security Tools
  Smart Card Support
  System Administration Tools
  System Management
Done
```

yum groupinfo-

```
[root@localhost adminuser]# yum groupsinfo "Development tools"
Loaded plugins: fastestmirror, langpacks
No such command: groupsinfo. Please use /usr/bin/yum --help
[root@localhost adminuser]# yum groupinfo "Development tools"
Loaded plugins: fastestmirror, langpacks
There is no installed groups file.
Maybe run: yum groups mark convert (see man yum)
Loading mirror speeds from cached hostfile
* base: mirrors.cat.pdx.edu
* extras: centos.den.host-engine.com
* updates: centos.unixheads.org

Group: Development Tools
Group-Id: development
Description: A basic development environment.
Mandatory Packages:
+autoconf
+automake
+binutils
+bison
+flex
+gcc
+gcc-c++
+gettext
+libtool
+make
+patch
+pkgconfig
+redhat-rpm-config
+rpm-build
+rpm-sign
Default Packages:
+byacc
+cscope
+ctags
+diffstat
+devtoolset-1
```

- Note- The command gives all the information of the group “Development tools”
- Also notice certain symbols like + - in front of package names.

yum groupinfo prefixes-

- 1. +(before the package name) – This depicts that the package can be installed with the next upgrade.
- 2. –(before the package name) – This depicts that the package is not and will not be installed as part of the group.
- 3. =(before the package name) – This depicts that the package was installed as part of the group.
- 4. blank space(before the package name)- This depicts that package was not installed as part of the group.

RPM commands with options, q= query message

Options	Descriptions-
rpm -ql	Lists all files that belong to a particular package
rpm -qf	Determines the package the file belongs to
rpm -qi	Provides information about the package
rpm -qc	Shows configuration files available
rpm -qd	Shows documentation available
rpm -V	Shows any parts of the package that has been changed since installation.
rpm -Va	Shows any parts of the package that has been changed since installation of all installed files.
rpm -qa	Shows all installed packages.

rpm -qa

```
[root@localhost adminuser]# rpm -qa
oddjob-0.31.5-4.el7.x86_64
grub2-2.02-0.34.el7.centos.x86_64
redland-1.0.16-6.el7.x86_64
lohit-marathi-fonts-2.5.3-2.el7.noarch
cups-libs-1.6.3-22.el7.x86_64
libgdata-0.17.1-1.el7.x86_64
poppler-data-0.4.6-3.el7.noarch
chrony-2.1.1-1.el7.centos.x86_64
firefox-45.3.0-1.el7.centos.x86_64
libetonyek-0.1.2-4.el7.x86_64
lohit-tamil-fonts-2.5.3-2.el7.noarch
gnome-icon-theme-3.12.0-1.el7.noarch
libgnomekbd-3.6.0-4.el7.x86_64
linux-firmware-20150904-43.git6ebf5d5.el7.noarch
screen-4.1.0-0.23.20120314git3c2946.el7_2.x86_64
liblangtag-0.5.4-8.el7.x86_64
rdate-1.4-25.el7.x86_64
libusb-0.1.6-3.el7.x86_64
evince-3.14.2-5.el7.x86_64
```

- This is just an extract of the entire output. Shows all installed packages.

rpm -ql kernel

```
[root@localhost adminuser]# rpm -ql kernel
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/acpi
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/acpi/acpi_extlog.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/acpi/acpi_ipmi.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/acpi/acpi_pad.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/acpi/apei
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/acpi/apei/einj.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/acpi/custom_method.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/acpi/ec_sys.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/acpi/sbs.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/acpi/sbsshc.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/acpi/video.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/ata
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/ata/acard-ahci.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/ata/ahci.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/ata/ahci_platform.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/ata/ata_generic.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/ata/ata_piix.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/ata/libahci.ko
/lib/modules/3.10.0-327.el7.x86_64/kernel/drivers/ata/libata.ko
```

- This is just an extract of the entire output.

rpm -qc

```
[root@localhost adminuser]# rpm -qc dnsmasq
/etc/dbus-1/system.d/dnsmasq.conf
/etc/dnsmasq.conf
[root@localhost adminuser]#
```

The above output shows 2 configuration files used by dnsmasq.

rpm -qd

```
[root@localhost adminuser]# rpm -qd dnsmasq
/usr/share/doc/dnsmasq-2.66/CHANGELOG
/usr/share/doc/dnsmasq-2.66/COPYING
/usr/share/doc/dnsmasq-2.66/DBus-interface
/usr/share/doc/dnsmasq-2.66/FAQ
/usr/share/doc/dnsmasq-2.66/doc.html
/usr/share/doc/dnsmasq-2.66/setup.html
/usr/share/man/man8/dnsmasq.8.gz
[root@localhost adminuser]#
```

package manager for Debian

- The package manager for Debian and distros similar to Debian, as Ubuntu, is called dpkg which stands for Debian package.
- Dpkg without anyone else oversees packages, can introduce from downloaded software package.
- Different devices like the apt and aptitude can look online repositories for packages and download packages and afterward hand them off to dpkg for management.

apt commands-

- apt stands for Advanced Packaging tool.
- apt –get remove packagename

```
ubuntu@ubuntu:~$ sudo apt-get remove thunderbird
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages will be REMOVED:
  thunderbird thunderbird-gnome-support thunderbird-locale-de thunderbird-locale-en thunderbird-locale-en-gb thunderbird-locale-en-us thunderbird-locale-es thunderbird-locale-es-ar
  thunderbird-locale-es-es thunderbird-locale-fr thunderbird-locale-it thunderbird-locale-pt thunderbird-locale-pt-br thunderbird-locale-pt-pt thunderbird-locale-ru thunderbird-locale-zh-cn
  thunderbird-locale-zh-hans thunderbird-locale-zh-hant thunderbird-locale-zh-tw
0 upgraded, 0 newly installed, 19 to remove and 0 not upgraded.
After this operation, 109 MB disk space will be freed.
Do you want to continue? [Y/n] y
(Reading database ... 193482 files and directories currently installed.)
Removing thunderbird-locale-zh-tw (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-zh-hant (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-zh-cn (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-zh-hans (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-gnome-support (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-de (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-en-gb (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-en-us (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-en (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-es-ar (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-es-es (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-es (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-fr (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-it (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-pt-br (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-pt-pt (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-pt (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird-locale-ru (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Removing thunderbird (1:45.7.0+build1-0ubuntu0.16.04.1) ...
Processing triggers for gnome-menus (3.13.3-6ubuntu3.1) ...
Processing triggers for desktop-file-utils (0.22-1ubuntu5) ...
Processing triggers for bamfdaemon (0.5.3-bzr0+16.04.20160824-0ubuntu1) ...
Rebuilding /usr/share/applications/bamf-2.index...
Processing triggers for mime-support (3.59ubuntu1) ...
Processing triggers for man-db (2.7.5-1) ...
ubuntu@ubuntu:~$
```

apt list --installed

```
ubuntu@ubuntu:~$ sudo apt list --installed
Listing... Done
a11y-profile-manager/xenial,now 0.1.10-0ubuntu3 amd64 [installed]
a11y-profile-manager-indicator/xenial,now 0.1.10-0ubuntu3 amd64 [installed]
account-plugin-facebook/xenial,now 0.12+16.04.20160126-0ubuntu1 all [installed]
account-plugin-flickr/xenial,now 0.12+16.04.20160126-0ubuntu1 all [installed]
account-plugin-google/xenial,now 0.12+16.04.20160126-0ubuntu1 all [installed]
accountsservice/xenial-updates,now 0.6.40-2ubuntu11.3 amd64 [installed]
acl/xenial,now 2.2.52-3 amd64 [installed]
acpi-support/xenial,now 0.142 amd64 [installed]
acpid/xenial,now 1:2.0.26-1ubuntu2 amd64 [installed]
activity-log-manager/xenial,now 0.9.7-0ubuntu23 amd64 [installed]
adduser/xenial,now 3.113+nmu3ubuntu4 all [installed]
adium-theme-ubuntu/xenial-updates,now 0.3.4-0ubuntu1.1 all [installed]
adwaita-icon-theme/xenial-updates,now 3.18.0-2ubuntu3.1 all [installed]
aisleriot/xenial,now 1:3.18.2-1ubuntu1 amd64 [installed]
alsa-base/xenial,now 1.0.25+dfsg-0ubuntu5 all [installed]
alsa-utils/xenial,now 1.1.0-0ubuntu5 amd64 [installed]
anacron/xenial,now 2.3-23 amd64 [installed]
apg/xenial,now 2.2.3.dfsg.1-2ubuntu1 amd64 [installed]
app-install-data/xenial,now 15.10 all [installed]
app-install-data-partner/xenial,now 16.04 all [installed]
apparmor/xenial-updates,now 2.10.95-0ubuntu2.5 amd64 [installed]
appmenu-qt/xenial,now 0.2.7+14.04.20140305-0ubuntu2 amd64 [installed]
appmenu-qt5/xenial-updates,now 0.3.0+16.04.20151130-0ubuntu2 amd64 [installed]
appport/xenial-updates,now 2.20.1-0ubuntu2.5 all [installed]
appport-gtk/xenial-updates,now 2.20.1-0ubuntu2.5 all [installed]
appport-symptoms/xenial,now 0.20 all [installed]
appstream/xenial-updates,now 0.9.4-1ubuntu2 amd64 [installed]
```

This command shows all the installed packages on the system.

dpkg commands-

- `dpkg -l` - This command lists all installed packages.

```
ubuntu@ubuntu:~$ dpkg -l
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/halF-conf/Half-inst/trig-aWait/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
||/ Name                               Version                               Architecture Description
+++-----+-----+-----+-----+
ii  a11y-profile-manager                 0.1.10-0ubuntu3                      amd64      Accessibility Profile Manager - Command-line utility
ii  a11y-profile-manager-indicator       0.1.10-0ubuntu3                      amd64      Accessibility Profile Manager - Unity desktop indicator
ii  account-plugin-facebook              0.12+16.04.20160126-0ubuntu         all        GNOME Control Center account plugin for single signon - facebook
ii  account-plugin-flickr               0.12+16.04.20160126-0ubuntu         all        GNOME Control Center account plugin for single signon - flickr
ii  account-plugin-google               0.12+16.04.20160126-0ubuntu         all        GNOME Control Center account plugin for single signon
ii  accountsservice                     0.6.40-2ubuntu11.3                  amd64      query and manipulate user account information
ii  acl                                  2.2.52-3                             amd64      Access control list utilities
ii  acpi-support                        0.142                                amd64      scripts for handling many ACPI events
ii  acpid                               1:2.0.26-1ubuntu2                   amd64      Advanced Configuration and Power Interface event daemon
ii  activity-log-manager                0.9.7-0ubuntu23                     amd64      blacklist configuration user interface for Zeitgeist
ii  adduser                             3.113+nmu3ubuntu4                   all        add and remove users and groups
ii  adium-theme-ubuntu                  0.3.4-0ubuntu1.1                    all        Adium message style for Ubuntu
ii  adwaita-icon-theme                  3.18.0-2ubuntu3.1                   all        default icon theme of GNOME (small subset)
ii  aisleriot                           1:3.18.2-1ubuntu1                   amd64      GNOME solitaire card game collection
ii  alsa-base                           1.0.25+dfsg-0ubuntu5                all        ALSA driver configuration files
ii  alsa-utils                          1.1.0-0ubuntu5                      amd64      Utilities for configuring and using ALSA
ii  anacron                             2.3-23                              amd64      cron-like program that doesn't go by time
ii  appg                                 2.2.3.dfsg.1-2ubuntu1               amd64      Automated Password Generator - Standalone version
ii  app-install-data                    15.10                                all        Ubuntu applications (data files)
ii  app-install-data-partner             16.04                                all        Application Installer (data files for partner applications/repositories)
ii  apparmor                             2.10.95-0ubuntu2.5                  amd64      user-space parser utility for AppArmor
ii  appmenu-qt:amd64                    0.2.7+14.04.20140305-0ubuntu        amd64      application menu for Qt
ii  appmenu-qt5                         0.3.0+16.04.20151130-0ubuntu        amd64      application menu for Qt5
ii  apport                               2.20.1-0ubuntu2.5                   all        automatically generate crash reports for debugging
ii  apport-gtk                          2.20.1-0ubuntu2.5                   all        GTK+ frontend for the apport crash report system
ii  apport-symptoms                     0.20                                all        symptom scripts for apport
ii  appstream                           0.9.4-1ubuntu2                      amd64      Software component index
ii  apt                                  1.2.19                              amd64      commandline package manager
ii  apt-clone                           0.4.1ubuntu1                        all        Script to create state bundles
ii  apt-transport-https                 1.2.19                              amd64      https download transport for APT
ii  apt-utils                           1.2.19                              amd64      package management related utility programs
ii  aptdaemon                           1.1.1+bzr982-0ubuntu14              all        transaction based package management service
ii  aptdaemon-data                      1.1.1+bzr982-0ubuntu14              all        data files for clients
```

dpkg -L packagename

```
ubuntu@ubuntu:~$ dpkg -L perl
./
/etc
/etc/perl
/etc/perl/Net
/etc/perl/Net/libnet.cfg
/etc/perl/CPAN
/usr
/usr/bin
/usr/bin/splain
/usr/bin/shasum
/usr/bin/pod2usage
/usr/bin/perlbug
/usr/bin/ptargrep
/usr/bin/ptar
/usr/bin/podchecker
/usr/bin/encguess
/usr/bin/pod2man
/usr/bin/prove
/usr/bin/pod2text
/usr/bin/h2xs
/usr/bin/c2ph
/usr/bin/pl2pm
/usr/bin/instmodsh
/usr/bin/piconv
/usr/bin/pstruct
/usr/bin/perlthanks
/usr/bin/h2ph
/usr/bin/xsubpp
/usr/bin/ptardiff
/usr/bin/libnetcfg
/usr/bin/zipdetails
/usr/bin/perlvp
/usr/bin/json_pp
/usr/bin/pod2html
/usr/bin/enc2xs
/usr/bin/cpan
/usr/bin/podselect
/usr/bin/corelist
/usr/bin/perldoc
/usr/bin/prename
```

- This command shows all the files present in that particular package.

Important Note: practice, practice,?

- Remember these are only few such commands discussed above.
- It is difficult to cover all commands, thus practice is required to get fluent with linux commands.

End of session